



# SYNTROPY

Don't manage time. Execute it.

## HACKATHON MVP IMPLEMENTATION PLAN

Created for: VibeCode 2.0 Hackathon

Platform: Next.js + Google Gemini 1.5 Pro + GCP

Generated: 2026-06-30

## [Award] Syntropy Hackathon MVP Implementation Plan

### NOTE

Building the "Autonomous Execution Partner" as specified in the [vibecode\_master\_blueprint.md](file:///c:/Users/KIIT/OneDrive/Documents/codingnijas/vibecode\_master\_blueprint.md)

## Scope

Per Part 16 of the blueprint, this implementation targets the **Hackathon MVP (24-48 hour scope)**:

1. [Completed] Google Auth login
2. [Completed] Read Google Calendar & Gmail (simulated with realistic mock data for demo)
3. [Completed] **The Wow Factor**: Anticipatory Execution engine AI auto-generates draft responses/summaries before tasks start
4. [Completed] Beautiful, fluid dark-mode UI

### NOTE

[!IMPORTANT]\*\* Gemini API & Google OAuth: \*\* Since this is a hackathon MVP, the AI features and Google integrations will use \*\*simulated/mock data\*\* with realistic animations to demonstrate the concept. This avoids requiring API keys and OAuth credentials during initial build. The architecture is designed so real APIs can be plugged in later.

## Open Questions

### NOTE

[!IMPORTANT] 1. \*\*Do you have Gemini API keys\*\* you'd like to integrate, or should I build with mock AI responses that demonstrate the concept? 2. \*\*Google OAuth credentials\*\* Do you have a Google Cloud project with OAuth configured, or should I build the auth flow with mock login? 3. \*\*Deployment target\*\* Should I set up Firebase Hosting deployment, or is local development sufficient for now?

## Proposed Changes

This will be a **Next.js 14** application with the full Syntropy experience.

## Project Structure

```
/syntropy-core
  /src
    /app                                # Next.js App Router
      layout.js                         # Root layout with dark theme
      page.js                           # Landing/Login page
    /dashboard
      page.js                           # Main Nexus Dashboard
      layout.js                         # Dashboard layout with sidebar
    /components
      /ui                               # Reusable UI primitives
        Button.js
        Card.js
        GlassPanel.js
        Badge.js
        Avatar.js
      /dashboard
        Timeline.js                    # Dynamic heartbeat timeline
        TaskCard.js                   # Individual task with risk glow
        BurnoutRadar.js               # Cognitive load heat map
        RiskIndicator.js              # Risk status visualization
        MorningBriefing.js            # AI morning briefing panel
        AnticipationEngine.js         # "Start for Me" feature
        NegotiationPanel.js          # Auto-negotiation UI
      /layout
        Sidebar.js                    # Navigation sidebar
        Header.js                     # Top bar with user info
      /effects
        AuroraEffect.js               # Pulsing aurora background
        ParticleField.js              # Ambient particle animation
    /lib
      mockData.js                     # Realistic calendar/email/task data
      aiEngine.js                     # Simulated AI anticipation engine
      animations.js                   # Framer Motion animation configs
    /styles
      globals.css                     # Design system & tokens
  package.json
  next.config.js
  README.md
```

## 1. Foundation & Design System

### [NEW] `package.json`

- Next.js 14, React 18, Framer Motion for animations
- Google Fonts (Inter/Geist)

### [NEW] `src/styles/globals.css`

- Dark mode design system with the specified palette:
  - Base black: `#0A0A0A`
  - Neon violet: `#8B5CF6`
  - Cyan accent: `#06B6D4`
- Glassmorphism utilities
- Custom animations (aurora pulse, glow effects, heartbeat)
- Typography system using Inter font

## 2. Visual Effects & Animations

### [NEW] `src/components/effects/AuroraEffect.js`

- Animated aurora borealis background using CSS gradients + Framer Motion
- Triggers when AI is "preparing" a task

### [NEW] `src/components/effects/ParticleField.js`

- Ambient floating particles for premium feel
- Subtle, non-distracting background effect

## 3. Core Dashboard Components

### [NEW] `src/components/dashboard/Timeline.js`

- Dynamic "heartbeat monitor" style timeline (not a boring list)
- Tasks plotted along time axis with risk-glow coloring
- Animated transitions when tasks are rescheduled

### [NEW] `src/components/dashboard/TaskCard.js`

- Glassmorphic task cards with:
  - Risk status glow (green amber red pulsing)
  - "Start for Me" button triggering AI preparation animation
  - AI-generated prep status indicator
  - Progress ring animation

### [NEW] `src/components/dashboard/BurnoutRadar.js`

- 5-day cognitive load heat map visualization
- Color-coded day blocks showing predicted stress levels
- Animated bar chart with smooth transitions

### [NEW] `src/components/dashboard/MorningBriefing.js`

- AI voice-style briefing panel
- Typewriter animation for AI messages
- Action buttons for AI suggestions (approve/modify/dismiss)

**[NEW] `src/components/dashboard/AnticipationEngine.js`**

- The "wow factor" shows AI auto-generating drafts
- Streaming text animation (simulating Gemini output)
- Before/after split view of task preparation
- Countdown timer showing "prepared X minutes before deadline"

**[NEW] `src/components/dashboard/NegotiationPanel.js`**

- AI-drafted negotiation emails
- Calendar conflict visualization
- One-click approve/send interface

---

## 4. Layout Components

**[NEW] `src/components/layout/Sidebar.js`**

- Sleek glassmorphic sidebar navigation
- Icons for: Dashboard, Tasks, Calendar, Negotiations, Settings
- Animated active state indicators

**[NEW] `src/components/layout/Header.js`**

- User avatar, notification bell, AI status indicator
- "Syntropy is working..." animation when AI is active

---

## 5. Data & AI Engine (Mock)

**[NEW] `src/lib/mockData.js`**

- Realistic tasks matching the blueprint demo script:
  - 3 PM meeting with "furious client" (urgent, AI-prepared)
  - 5 PM coding assignment (at risk, negotiation candidate)
  - Morning standup (low risk, context prepared)
  - Additional tasks for visual richness
- Mock calendar events, email threads, AI-generated drafts

**[NEW] `src/lib/aiEngine.js`**

- Simulated AI responses with realistic delays

- "Anticipatory Execution" logic: pre-generates content for upcoming tasks
- Streaming text simulation for the "Start for Me" experience

---

## 6. Pages

### [NEW] `src/app/page.js` (Landing)

- Stunning dark landing page with the tagline "Don't manage time. Execute it."
- Animated hero with aurora effect
- Mock Google Sign-In button routes to dashboard

### [NEW] `src/app/dashboard/page.js` (The Nexus)

- Full dashboard assembling all components
- Real-time-feeling updates via state management
- The complete "Golden Path" demo experience

---

## 7. README

### [NEW] `README.md`

- Polished with badges, architecture diagram, setup instructions
- Screenshots section (populated after build)

---

## Verification Plan

### Automated Tests

- `npm run build` Verify production build succeeds with no errors
- `npm run dev` Visual verification of all components

### Manual Verification

- Landing page renders with full aurora animation
- Google Auth mock flow navigates to dashboard
- Dashboard shows timeline with risk-colored tasks
- "Start for Me" button triggers AI preparation animation with streaming text
- Burnout Radar displays 5-day heat map
- Negotiation panel shows draft email
- All micro-animations are smooth (60fps)
- Mobile responsive layout works

## Implementation Order

1. **Phase 1:** Project setup, design system, global styles
2. **Phase 2:** Visual effects (Aurora, Particles)
3. **Phase 3:** Layout components (Sidebar, Header)
4. **Phase 4:** Mock data & AI engine
5. **Phase 5:** Core dashboard components (Timeline, TaskCard, BurnoutRadar)
6. **Phase 6:** Wow features (AnticipationEngine, NegotiationPanel, MorningBriefing)
7. **Phase 7:** Pages assembly (Landing + Dashboard)
8. **Phase 8:** Polish, README, final verification